

Implementing Digital Filters and DSP Micro-Controller for Estimating the Frequency of a Time-Domain Signal

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Abstract—Accurate frequency estimation plays an important role in many applications, like power grids, micro-grid integration etc. This paper aims to estimate the frequency of a voltage signal at a certain frequency, with the help of another signal. The frequency of the other signal is kept close to the signal under test. The product is obtained, so as to get two components, the difference frequency component and the sum frequency component. This paper deals with the difference and the sum frequency component obtained, so as to get estimates of the frequency of the voltage signal source. To achieve this, a Low-Pass (LP) digital filter extracts the difference frequency signal and a Band-Pass (BP) digital filter extracts the sum frequency signal. Both are Butter-worth type of filters. This approach can also be used to analyze a power system waveform/signal that contains harmonics. This paper uses Texas Instruments' TMS320F28379d micro-controller unit to extract the difference and sum frequencies of the two sources. For further validation of the approach, MATLAB/Simulink based software simulations are also provided. Digital filters design is also discussed briefly.

Index Terms—MATLAB, Simulink, Digital Signal Processing (DSP), Digital filters, TI Microcontroller

I. INTRODUCTION

Texas Instruments' TMS320F28379d (see Appendix) is a powerful 32-bit floating-point micro-controller unit (MCU) designed for advanced real time applications. It supports a new dual-core C28x architecture that significantly boosts system performance. The integrated analog and control peripherals also let designers consolidate control architectures and eliminate multiprocessor use in high-end systems. It runs on C/C++ programming language, that is burned on the TMS320F28379d chip, using Code Composer Studio (CCS). Code Composer Studio is an integrated development environment (IDE) that supports TI's Micro controller and Embedded Processors portfolio. Code Composer Studio comprises a suite of tools used to develop and debug embedded applications [1].

Designing and debugging a code in CCS is sometimes hard and time consuming. MATLAB Simulink Coder along with Embedded coder and TI support package for C2000 provide an alternative and a powerful tool to program TI's micro controller units.

In [2], a DSP based watt-hour meter was designed, where line currents and voltages signals were acquired to measure real and reactive power. The author used 0.35 μm CMOS technology based IC to design high accuracy DSP based

watt-hour meter. Along the same lines, [3] derives using two standard algorithms: FFT and Goertzel. The algorithm was implemented in C language on a DSP microprocessor TMS 32010. [4] - [5], explains the use of DSP processors for power quality measurements. [6], author have used slow sampling rate to estimate the system frequency, using digital signal processing strategies. In [7] illustrates the procedure of estimating frequency of a signal using DFT coefficients. In the future it may be possible to map the approach discussed here to a custom VLSI chip [8].

The underlying motivation of this research is to estimate the frequency of the source voltage signal using digital filter and cost effective DSP micro-controllers. In this paper, TI MCU is used to multiply two voltage signals (at two different frequencies). For one of the signal the frequency is known, f_1 . The signal under test is supplied from function generator, and has frequency f_2 .

The product of the two signals is passed through low pass filter and separately through band pass filter, so as to separate the two components. The difference frequency component is observed at the oscilloscope and analyzed. The two filters are designed and the method is implemented using DSP micro-controller. Further validation is done with results obtained from MATLAB/Simulink based simulations.

The rest of paper is structured as follows. Section II introduces TI TMS320F28379D architecture and programming and Section III the mathematical expressions and notations used in the paper. Section IV describes the digital filter design. Section V explains the hardware implementations and results obtained. Section VI presents Simulink results. Section VII displays the comparison of the results obtained from hardware setup and simulink based simulations. Section VIII concludes the paper.

II. TMS320F28379D ARCHITECTURE AND PROGRAMMING

The TMS320F2837xD is a powerful 32-bit floating-point micro controller unit (MCU) designed for advanced closed-loop control applications such as industrial motor drives, solar inverters and digital power, electrical vehicles and transportation, and sensing and signal processing. The pin diagram of the TI MCU is shown in Fig. 1. To program the TI MCU, MATLAB Embedded coder along with Texas Instruments

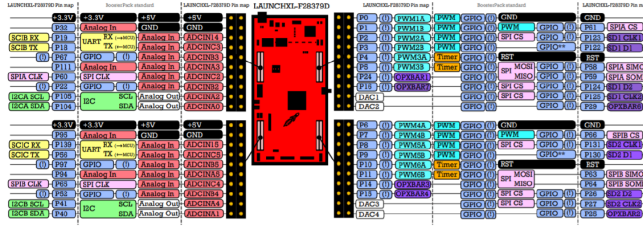


Fig. 1: Pin Diagram of TMS320F28379d Launchpad.

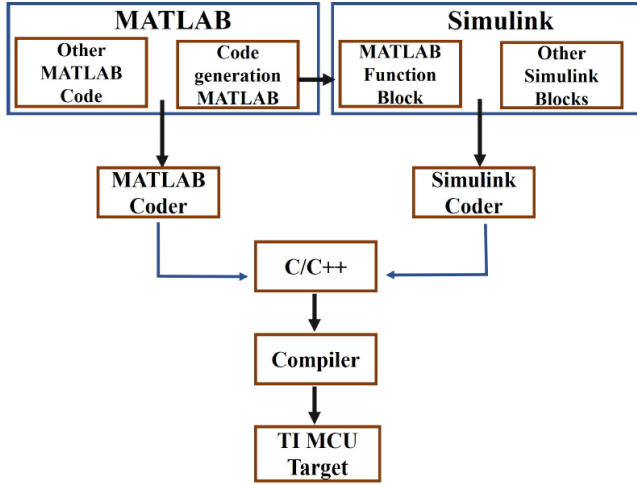


Fig. 2: Flowchart for programming TMS320F28379d Launchpad using MATLAB Embedded Coder.

support package was used. The flow chart of the programming is shown in Fig 2.

III. MATHEMATICAL EXPRESSIONS

Assuming, $v_1(t)$ and $v_2(t)$ are given by:

$$v_1(t) = V_{m1} \sin(\omega_1 t) \quad (1)$$

$$v_2(t) = V_{m2} \sin(\omega_2 t) \quad (2)$$

The product of $v_1(t)$ and $v_2(t)$, is represented by $z(t)$:

$$z(t) = v_1(t) \times v_2(t) \quad (3)$$

Using trigonometric identity :

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a - b) - \cos(a + b)] \quad (4)$$

$$z(t) = \frac{V_{m1} V_{m2}}{2} [\cos(\underbrace{(\omega_1 - \omega_2)t}_{\text{Difference Frequency}}) - \cos(\underbrace{(\omega_1 + \omega_2)t}_{\text{Sum Frequency}})] \quad (5)$$

It is observed from (5), that the product has two components, one difference frequency component, and the other the sum frequency. Notating, $x_1(t)$ as difference frequency component and $x_2(t)$ as sum frequency component.

A simple block diagram represents the method in Fig. 6.

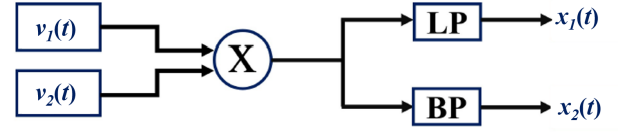


Fig. 3: Block Diagram of proposed method.

IV. FILTER DESIGN

The LP and BP filters shown in Fig. 6 were the result of comprehensive studies. These studies involved MATLAB simulations from the following perspectives:

- Effect of target bandwidth chosen (typically four cases were considered).
- Effect of filter order chosen, from the performance and complexity perspectives.
- Effect of the target frequency chosen for the BP filter, for example 120 Hz, or 119 Hz, or 115 Hz.

These studies involve both frequency domain and time domain evaluations. One more important point: The sampling frequency was chosen to be 10 kHz so that higher harmonics, such as the 9th, or 11th, could be studied conveniently, without changing the sampling frequency.

Indeed, we have performed experimentation on 9th harmonic detection. Naturally, f_2 was chosen to be $9 \times 60 = 540$ Hz. For $f_1 = 59.8$ Hz, we observed the frequency of the LP filter output to range from 1.75 Hz to 1.8 Hz, the ideal being 1.8 Hz. For $f_1 = 59$ Hz, we saw the output of the LP filter frequency to range from 8.77 Hz to 9.46 Hz; the ideal being 9 Hz. Better estimates could of course be obtained through averaging and/or by using more output cycles.

V. HARDWARE IMPLEMENTATION

The block diagram representation of the hardware setup to implemented to analyze the difference frequency component of $z(t)$ is shown in Fig. 4. Fig. 5 depicts the actual setup used in the lab. Here the TI MCU used TMS320F28379d, which is 32-bit floating point processor, with 12 bit ADCs on board. The clock frequency used for the DSP processor is 200 MHz. The function generator provides the current signal under test $v_2(t)$. The other signal $v_1(t)$, is generated within the MATLAB/Simulink model using a waveform generator.

The TI MCU is programmed using MATLAB's Embedded Coder app. The MATLAB/Simulink model implemented using Embedded coder is shown in Fig. 6. The test-setup was tested for three cases; *Case 1*: frequency of signal under test is 55 Hz, *Case 2*: frequency of signal under test is 59 Hz, and *Case 3*: frequency of signal under test is 59.8 Hz. Only the difference frequency component, i.e., $x_1(t)$ was observed on the oscilloscope, which was obtained from the DAC of the micro-controller. The peak to peak value of $v_1(t)$ and $v_2(t)$ are kept to be similar at 1V, with an offset of 0.5VDC. The results were observed on the oscilloscope and are presented in Fig. 7 - Fig. 9.

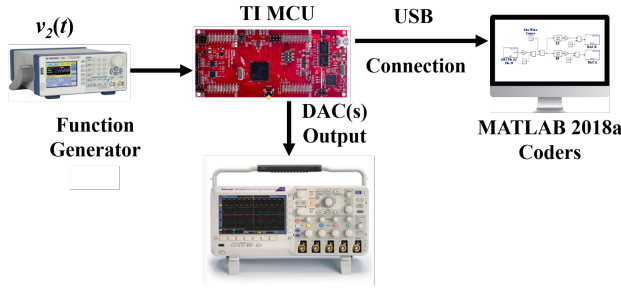


Fig. 4: Block diagram representation of the hardware setup with TMS320F28379d Launchpad.



Fig. 5: Actual hardware setup with TMS320F28379d Launchpad.

A few more test cases for done with decreasing the peak-peak value of the signal $v_2(t)$ to 0.5V, keeping the peak-peak value of $v_1(t)$ same at 1V. It can be observed from the Fig. 7- Fig. 9, that Low pass filter is providing us the difference frequency component. With this measurement, the frequency of the signal $v_2(t)$ can be estimated.

Remark: The voltage magnitude applied to the DSP micro-controller is well within the maximum range (0-3.3 V) set by the DSP controller.

VI. SIMULINK MODEL

To validate the Hardware results, a similar setup was modeled in MATLAB/Simulink. The block diagram is shown in Fig. 10. In the model saturation blocks are used to make the system closer to the hardware limitations. The results obtained for the three cases as discussed in previous sections are shown in Fig. 11, only signal $x_1(t)$ is presented. FFT was also performed to obtained the frequency spectrum of the $v_1(t)$, $v_2(t)$ and $x_1(t)$. The results are shown in Fig. 12 - Fig. 14. From Fig. 11 - Fig. 14, it is concluded that Low pass filter provides us the difference frequency component.

TABLE I: Parameters for Hardware Implementation

S.No	Parameter	Value
1	v_2	1 V _{pp} , 0.5 V _{DC} offset
2	f_1	60 Hz
3	N	1
4	Band Pass Filter	Num = [0.001443 0 -0.001443] Den = [1 -1.9919 0.99711]
5	Low Pass Filter	Num = [0.006244 0.006244] Den = [1 -1.9893 0.98939]
6	Sampling Frequency	10 kHz

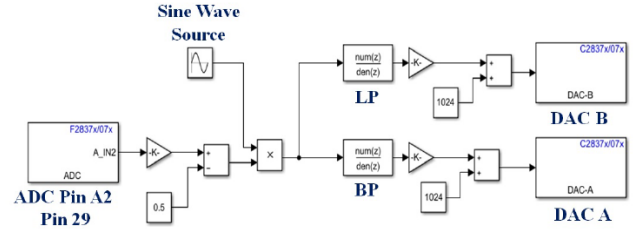


Fig. 6: MATLAB/Simulink model implemented on DSP microcontroller using Embedded coder.

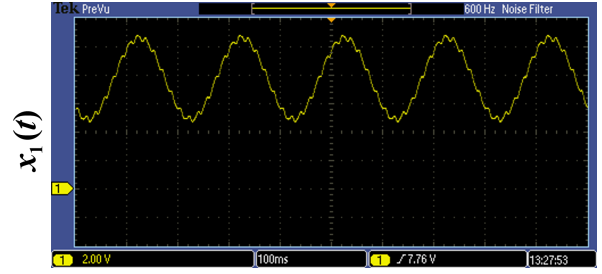


Fig. 7: Case 1: Signal $x_1(t)$ when the frequency of $v_2(t)$, i.e., $f_2 = 55\text{Hz}$. The time scale per division is 100 ms, and the frequency of $x_1(t) = 5\text{Hz}$

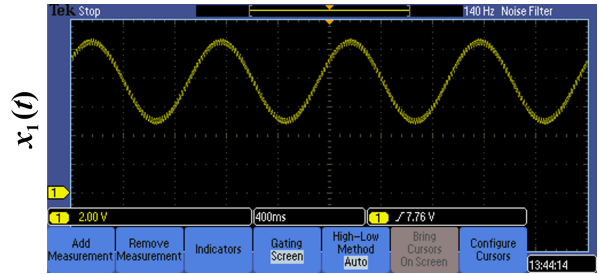


Fig. 8: Case 2: Signal $x_1(t)$ when the frequency of $v_2(t)$, i.e., $f_2 = 59\text{Hz}$. The time scale per division is 400 ms, and the frequency of $x_1(t) = 1\text{Hz}$.

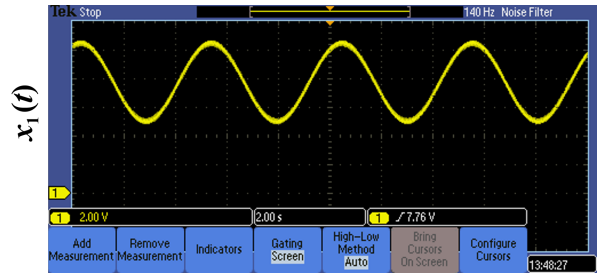


Fig. 9: Case 3: Signal $x_1(t)$ when the frequency of $v_2(t)$, i.e., $f_2 = 59.8\text{Hz}$. The time scale per division is 2s, and the frequency of $x_1(t) = 0.2\text{Hz}$.

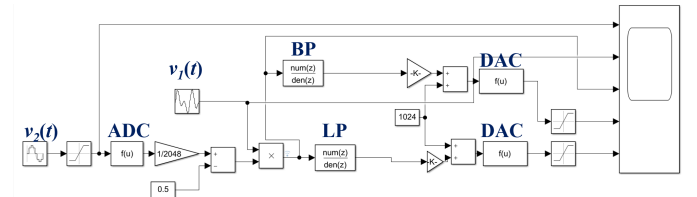


Fig. 10: Simulation model for validating the hardware test-bed.

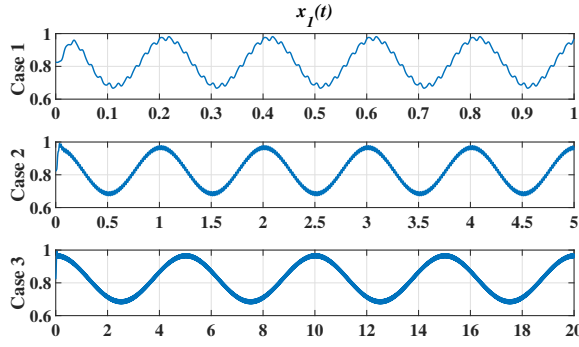


Fig. 11: Simulation results obtained for the three cases. For simplicity, only $x_1(t)$ signal is presented.

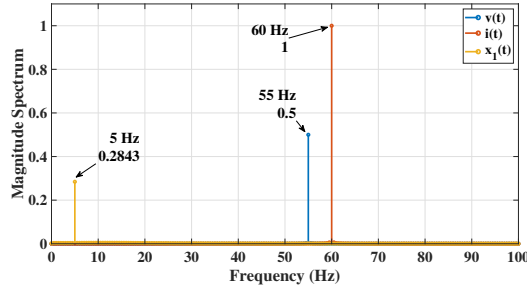


Fig. 12: Magnitude spectrum obtained from FFT for Case 1.

TABLE II: Comparison of peak-peak value of $x_1(t)$ obtained from Simulink model and hardware setup.

Mag of $v_2(t)$	f_1 (Hz)	Simulink Model	Hardware setup
0.5 V	55 Hz	0.154	0.31
	59 Hz	0.167	0.30
	59.8	0.178	0.29
1 V	55	0.31	0.62
	59	0.305	0.61
	59.8	0.306	0.59

VII. COMPARISON BETWEEN SIMULINK MODEL AND HARDWARE SETUP

For further investigation, and comparison, the magnitude of the signal $v_2(t)$ was changed, and the peak-peak to value of the signal $x_2(t)$ was observe, both from Simulink model and hardware setup. The results are tabulated in Table II. It is observed from Table II that as the peak to peak magnitude of $v_2(t)$ is increased by a factor of 2, peak to peak magnitude value of $x_1(t)$ is also increased in the same proportion. Furthermore, the ratio between results obtained from simulink model and hardware setup is approximately 2, and is consistent in all the cases considered.

VIII. CONCLUSION

A method was proposed to estimate the frequency of a time domain signal using digital filters and DSP micro-controllers. A signal whose frequency is known is multiplied with the signal under test. The product resulted was a combination of two components, one difference frequency component and the other sum frequency component. The product was passed

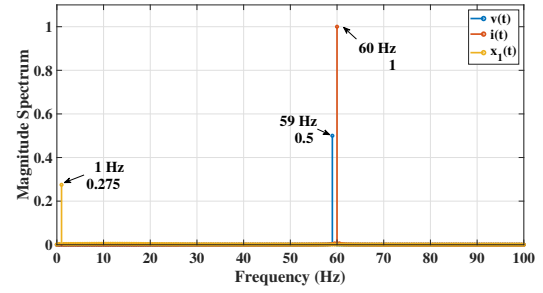


Fig. 13: Magnitude spectrum obtained from FFT for Case 2.

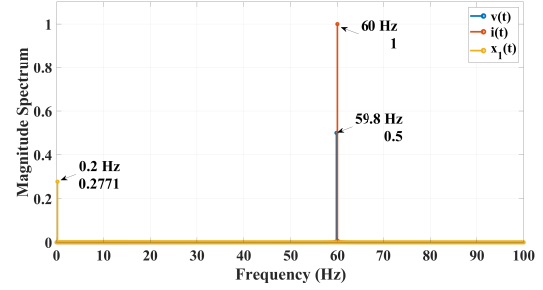


Fig. 14: Magnitude spectrum obtained from FFT for Case 3.

through two digital filters, Low pass filter and Band pass filter to separate the two components. The difference frequency component was studied to estimate the frequency of signal under test. The hardware setup was replicated using MATLAB/Simulink for validation of the proposed method. The results and observations were recorded and analyzed. Future work will consist of extension of the proposed method to estimate the frequency of the signal in a three-phase network.

APPENDIX

The MCU TMS320F28379d Launchpad used in this paper is shown in Fig. 15.

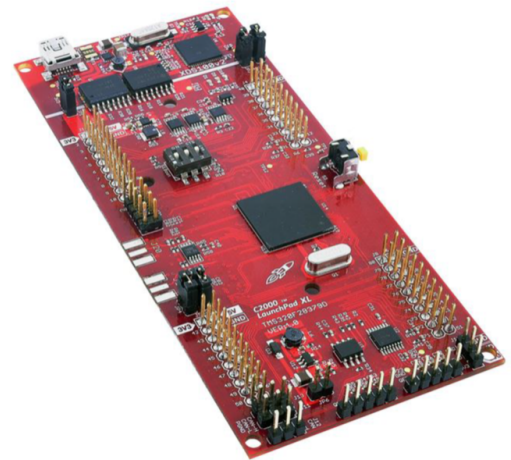


Fig. 15: Texas Instrument's C2000 TMS320F28379d Launchpad.

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